

Operation Timing and 30-Day Mortality After Elective General Surgery

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BACKGROUND: Human factors such as fatigue, circadian rhythms, scheduling, and staffing may have an impact on patient care over the course of a day across all medical specialties. Research by the transportation industry concludes that human performance is degraded by shift work, circadian rhythm disturbances, and prolonged duty. We investigated whether the timing of general surgery (specifically, increasing time of day, increasing day of week, July/August cases versus other months), and moon phase is independently related to 30-day mortality. A secondary outcome of composite in-hospital complications was also evaluated.

METHODS: The binary outcomes of 32,001 elective general surgical patients at the Cleveland Clinic between January 2005 and September 2010 were analyzed according to the hour of the day (6 am to 7 pm), day of the workweek, month of the year, and moon phase in which the surgery started. Thirty-day mortality was modeled as a binary endpoint using a multivariable logistic regression, adjusting for a risk stratification index based on International Classification of Diseases (9th rev.) codes.

RESULTS: The adjusted odds ratio ([Bonferroni-adjusted 95% CI]) associated with a relative increase in time of day of 4 h was 1.23 [0.91, 1.67], $P = 0.09$. Similarly, no association was found for day of week (0.99 [0.83, 1.17]) for a relative increase of 1 day, $P = 0.85$. Mortality was not significantly more frequent in July and August than in other months (adjusted odds ratio = 0.72 [0.36, 1.43], $P = 0.22$). Moon phase was not significantly related to mortality ($P = 0.72$). There were also no significant time-dependent differences in composite complications.

CONCLUSIONS: Elective general surgery appears to be comparably safe at any time of the workday, any day of the workweek, and in any month of the year. (Anesth Analg 2011;113:1423–8)

It is well established that inadequate sleep, whether from prolonged duty or circadian rhythm disturbances, degrades performance.^{1,2} A consequence is that fatigue is thought to contribute to 15%–20% of all transportation accidents.² Because there is no reason to assume that hospital personnel are immune to the performance-degrading effects of sleep deprivation,³ resident work hours are increasingly being restricted to reduce fatigue and the potential for related errors.⁴

Even excluding the obvious sleep deprivation associated with overnight work,^{5,6} hospital personnel are likely to become progressively fatigued and work less effectively during the course of a normal workday.^{5,6} Anesthesiologists may be at particular risk because prolonged monitoring is especially impaired by fatigue.⁷ It is similarly likely that hospital personnel become progressively fatigued as the normal workweek progresses from Monday to Friday.⁸

An additional time-related factor that might influence clinician performance is that most new residents enter teaching hospitals in July and August, and the responsibilities of existing residents often precipitously increase at the same time.

Long learning curves associated with anesthesia and surgical procedures may increase risks in the operating rooms during these months^{9,10} and therefore worsen patient outcomes.¹¹

We therefore tested the hypotheses that the risk of 30-day mortality associated with elective general surgery: 1) increases from morning to evening throughout the routine workday; 2) increases from Monday to Friday through the workweek; and 3) is more frequent in July and August than during other months of the year. As a presumed negative control, we also evaluated mortality as a function of the phase of the moon. Secondarily, we evaluated these hypotheses as they pertain to a composite in-hospital morbidity endpoint. The analysis was restricted to elective operations, because urgent or semiurgent procedures, which are inherently riskier, are often performed later in the workday even without being specifically labeled as “emergencies.”

METHODS

Use of our Perioperative Health Documentation System was approved by the IRB, Cleveland Clinic, Cleveland, Ohio. This registry includes the entire electronic anesthesia record and data from various administrative databases. Perioperative variables were prospectively collected concurrently with patient care from our electronic anesthesia record and other electronic systems. Only adults managed by the Department of General Anesthesia were included, but monitored anesthesia care, neuraxial anesthesia, nerve blocks, general anesthesia, and combinations thereof were allowed. Elective general surgical procedures are rarely performed in the evenings or weekends at the Cleveland Clinic; evening and weekend cases were thus excluded from our analysis. No obstetrical care is provided at the Cleveland Clinic Main Campus.

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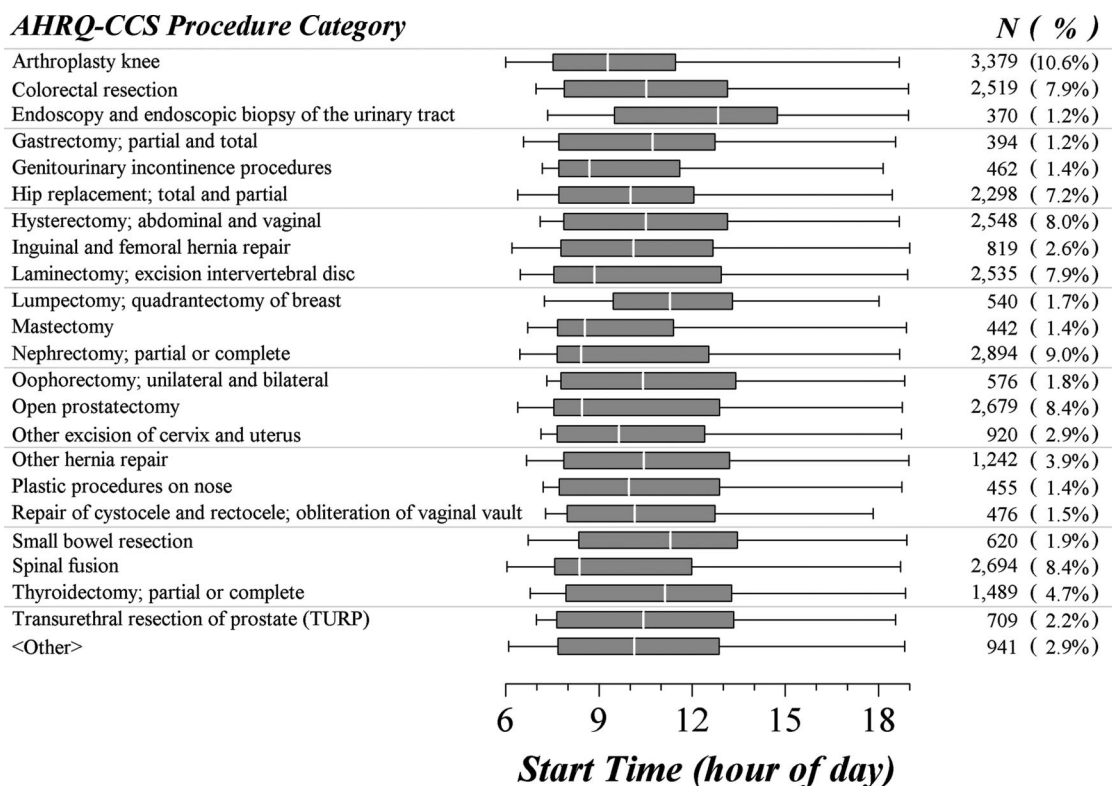


Figure 1. Boxplots of case start times, given by the United States Agency for Healthcare Research and Quality's single-level Clinical Classifications Software for Procedures (AHRQ-CCS). Gray boxes are drawn between the first and third quartiles of the distribution, a white mark is drawn at the median, and "whiskers" extend from the quartiles to either the maximum or minimum of the distribution.

Most elective general surgical cases are scheduled to begin at 07:30 and finish by 19:00. Most cases at the Cleveland Clinic Main Campus are performed in their originally scheduled operating rooms, and in the designated order. However, cases are moved from one operating room to another when doing so improves efficiency. Emergency and urgent cases are, of course, done as necessary and will occasionally take precedence over scheduled cases. However, the Main Campus is not a trauma center and emergent cases constitute only 4.8% of the overall surgical caseload.

It is impossible for us to determine from our registry which cases might have been urgent, but not coded as emergencies. The distinction is important, though, as urgent cases (with their presumably higher risk) are most often performed after scheduled cases. We thus sought to analyze only certain types of procedures for which elective status is virtually certain. To filter our registry for such elective cases, we used the United States Agency for Healthcare Research and Quality's Clinical Classifications Software (AHRQ-CCS) for the International Classification of Diseases, 9th revision, Clinical Modification (ICD-9-CM). The AHRQ-CCS system is a data aggregation tool that collapses >14,000 ICD-9-CM diagnosis codes and 3900 ICD-9-CM procedure codes into a smaller number of clinically meaningful categories.* AHRQ-CCS procedure categories representing the elective surgeries under consideration for our study are thus given in Figure 1.

*Source: <http://www.hcup-us.ahrq.gov/toolssoftware/ccs/ccs.jsp>; accessed 07/16/2010.

88,088 General surgical patients treated between January 2005 and September 2010 at Cleveland Clinic

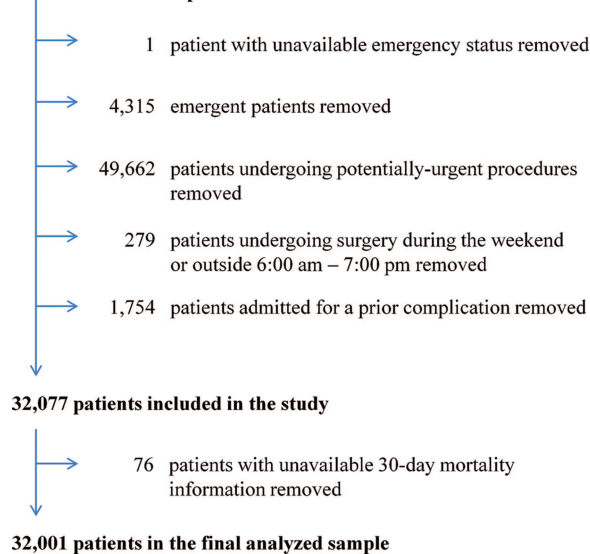


Figure 2. Summary of inclusion and exclusion criteria.

Excluded were weekend and overnight cases (i.e., between 19:00 and 05:59), patients admitted as a result of a complication from a previous surgery (based on matching of medical record numbers), and patients in whom 30-day mortality was missing. A summary of inclusion and exclusion criteria for our study is provided in Figure 2.

The primary outcome of our study was all-cause 30-day mortality, which was obtained from a review of hospital records and a search of the Social Security Death Index. The Death Index is reliable, especially since approximately a year elapsed between enrollment of the last patient and our analysis. Our secondary endpoint was a composite of complications defined by AHRQ-CCS diagnosis categories 237 (complication of device; implant or graft) and 238 (complications of surgical procedures or medical care).

The exposures of interest (time of day, day of week, month of year, and phase of moon) were based on surgical start times. Phase of moon time stamps were obtained from the United States Naval Observatory, Washington, DC; with each patient, we associated the moon phase for which the reported time stamp (time stamps adjusted from Greenwich Mean Time to Eastern Standard Time, considering daylight savings time) was nearest to the patient's surgical start time. Because we expected there to be no relationship between phase of moon and our primary outcomes, this exposure served as a negative control.

Statistical Analysis

R software version 2.12.1 for 64-bit Windows (The R Foundation for Statistical Computing, Vienna, Austria) was used for all statistical analysis.

Thirty-day mortality was modeled as a binary endpoint using multivariable logistic regression. We adjusted for diagnoses and procedures using the Risk Stratification Index (RSI) for 30-day mortality.¹² RSI is an accurate system (C-statistic 0.85) for predicting 30-day mortality from ICD-9-CM codes. Complications, as defined by AHRQ-CCS, were omitted from the model. In addition to adjusting for RSI, we also adjusted for the overall incidence of 30-day mortality for each AHRQ-CCS procedure category; this was done to ensure adequate adjustment for potential confounding associated with any differences in the exposures of interest with respect to type of procedure.

First, as an exploratory analysis, we developed a model that estimated the adjusted incidence of mortality as a function of the 4 exposures of interest; for this model, time of day and day of week were analyzed via smoothing (specifically, with restricted cubic splines,¹³ while the adjusted mortality incidence was estimated separately for each month and for each moon phase).

For our primary model, we assumed linearity in the relationships for time of day and day of week and reported odds ratios corresponding to relative increases in each factor. Also, month was coded as a 2-level factor (July/August versus other months) to test the "residency effect" hypothesis. Within this primary model we also evaluated the hypothesis that odds of 30-day mortality is in general associated with moon phase.

The Wald test for generalized linear model parameters was used to evaluate these 4 hypotheses; a Bonferroni-adjusted significance criterion of 0.0125 was used for these tests to maintain an overall false positive (Type I) error rate of 0.05 for the primary analysis. The secondary endpoint of composite in-hospital morbidity was analyzed with the same approach.

We had >90% power to detect the following odds ratios: 1.2 or larger for a 4-h increase in time of day, 1.2 or larger

Table 1. Summary of Baseline Risk Factors for 32,001 General Surgical Patients

Factor	Statistics*
Age (y)	57.7 ± 15.1
Male gender	14,768 (46.2)
Race	
Caucasian	26,488 (84.0)
African American	3,790 (12.0)
Other	1,243 (3.9)
ASA physical status	
I-II	17,261 (54.0)
III	13,677 (42.7)
IV-VI	1,055 (3.3)
Body mass index (kg/m ²)	28.2 [24.6, 32.8]
Cancer	10,958 (34.2)
Cardiovascular/cerebrovascular disease	16,176 (50.5)
Dementia	242 (0.8)
Diabetes	4,166 (13.0)
Digestive disease	7,037 (22.0)
Osteoarthritis	5,719 (17.9)
Psychiatric disorder	2,910 (9.1)
Pulmonary disease	3,493 (10.9)
Charlson Comorbidity Index	0.0 [0.0, 2.0]
Risk Stratification Index (30-day mortality)	-0.3 [-1.2, 0.0]
Risk Stratification Index (in-hospital complications)	-0.3 [-0.8, 0.0]

*Statistics given by means ± SDs, N (percent), or median [first quartile, third quartile].

Table 2. Frequencies of Case Start Times by Hour of Day, Day of Week, Month of Year, and Moon Phase

N (%) of cases		N (%) of cases	
Factor operation hour		Factor month	
6:00	562 (1.8)	January	2,670 (8.3)
7:00	10,631 (33.2)	February	2,506 (7.8)
8:00	3,807 (11.9)	March	2,697 (8.4)
9:00	1,664 (5.2)	April	2,698 (8.4)
10:00	2,501 (7.8)	May	2,654 (8.3)
11:00	2,855 (8.9)	June	2,994 (9.4)
12:00	2,763 (8.6)	July	2,325 (7.3)
13:00	2,623 (8.2)	August	3,177 (9.9)
14:00	2,063 (6.4)	September	3,208 (10.0)
15:00	1,267 (4.0)	October	2,689 (8.4)
16:00	745 (2.3)	November	2,544 (7.9)
17:00	356 (1.1)	December	1,839 (5.7)
18:00	164 (0.5)		
Day of week		Phase of moon	
Monday	7,005 (21.9)	New Moon	7,708 (24.1)
Tuesday	7,008 (21.9)	First Quarter	8,100 (25.3)
Wednesday	6,266 (19.6)	Full Moon	8,051 (25.2)
Thursday	5,635 (17.6)	Last Quarter	8,142 (25.4)
Friday	6,087 (19.0)		

for a 1-day increase in day of week, and 2.0 or larger when comparing patients treated in July or August to patients treated in other months. Though an odds ratio of 2.0 is ostensibly large, it represents an approximate change in incidence from the observed overall 30-day mortality incidence of 0.43% to 0.86%. For the composite complication outcome, the odds ratios for which we had >80% power were, respectively, 1.13, 1.12, and 1.17, for the effects described above.

RESULTS

Baseline demographics and morphometrics for the 32,001 elective general surgery patients included in our study are

Adjusted Incidence of 30-Day Mortality

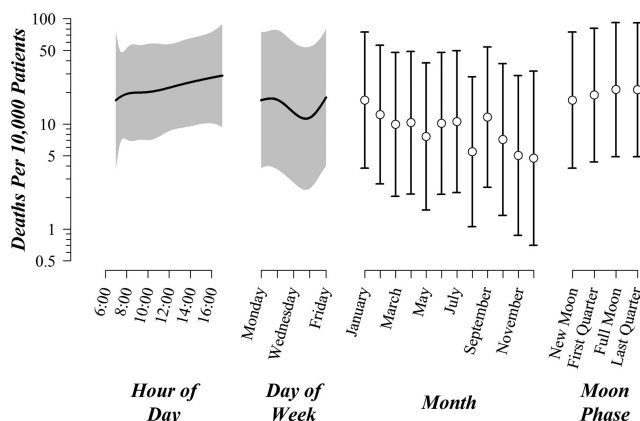


Figure 3. Adjusted probability estimates of 30-day mortality, and their associated 98.75% confidence intervals, by hour of day, day of week, month, and moon phase. The predicted incidence of 30-day mortality is generally lower for these adjusted estimates primarily due to the fact that the mean Risk Stratification Index of -0.53 was associated with an approximately 0.053% incidence of 30-day mortality.

Adjusted Incidence of In-Hospital Complications

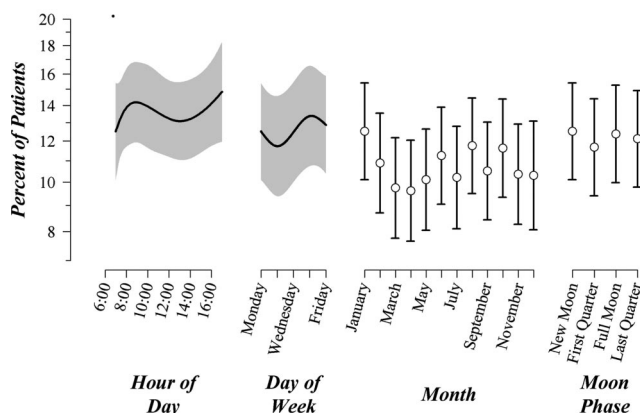


Figure 4. Adjusted probability estimates of a composite outcome comprised of Agency for Healthcare Research and Quality-defined complications, and their associated 98.75% confidence intervals, by hour of day, day of week, month, and moon phase.

summarized in Table 1, and the frequency distributions of hour of day, day of week, month, and moon phase associated with the patients' surgical start times are given in Table 2.

The crude/unadjusted 30-day mortality incidence [95% confidence interval (CI)] was 0.43% [0.36% , 0.51%]; RSI and AHRQ-CCS procedure-category-adjusted estimates of this incidence were similar across the 4 factors of interest (Fig. 3). The adjusted odds ratio (AOR) [Bonferroni-adjusted 95% CI] associated with a relative increase in time of day of 4 h was 1.23 [0.91 , 1.67], which was not different from 1.0 ($P = 0.09$). Similarly, no association was found for day of week ($P = 0.85$; AOR = 0.99 [0.83 , 1.17] for a relative increase of 1 day). Thirty-day mortality incidence of July/August cases was not significantly different from mortality in surgical cases occurring in other months ($P = 0.22$; AOR = 0.72 [0.36 , 1.43], July/August versus other

months). As expected, moon phase was not significantly related to adjusted odds of 30-day mortality ($P = 0.72$ for an overall test of difference among the 4 lunar phases).

For the secondary outcome of composite (AHRQ-defined) in-hospital complications, we estimated the crude/unadjusted incidence to be 13.3% [13.0% , 13.7%]. RSI-adjusted incidence estimates as they relate to time of day, day of week, month, and moon phase are shown in Figure 4. Hour of day ($P = 0.73$, AOR = 1.01 [0.95 , 1.07] for a relative increase of 4 h), day of week ($P = 0.09$, AOR = 1.02 [0.99 , 1.05] for a relative increase of 1 day), month ($P = 0.36$ comparing July/August cases to cases occurring in other months, AOR = 1.04 [0.93 , 1.17]), and moon phase ($P = 0.41$ for an overall test of difference among the four phases) were all not significantly related to the incidence of the composite complication outcome.

DISCUSSION

Comparable mortality throughout the workday among patients having general surgery is consistent with recent findings in cardiac surgery¹⁴ and heart transplant¹⁵ patients, in which later surgical start times were not associated with increased mortality. However, our results contrast with Kelz, who evaluated patients in the Veterans Administration system and observed a 25% mortality risk increase when 16:00 to 18:00 starts were compared to operations starting between 07:00 and 16:00.¹⁶ Our results also contrast with Wright et al., who found that the probability of anesthetic-related adverse events increased from a low of 1.0% at 09:00 to 4.2% at 16:00.¹⁷

That Kelz and Wright et al. found mortality to be worse when elective general surgery started later in the workday does not prove that the association is causal, because selection bias may contribute to the apparent worsening of mortality late in the workday.^{16,17} For example, there is a class of urgent or semiurgent patients between truly elective cases and those that are declared to be emergencies. This distinction is important because urgent or semiurgent status is not recorded in registries or administrative databases, but these patients are presumably sicker and, because of time constraints, may not get extensive preoperative evaluations. Many of these operations will be performed after completion of the routine elective cases, that is, later in the day. To the extent that these patients are sicker or less well prepared for surgery, including them in the analysis will artifactually make surgery performed later in the day look worse. This particular bias will be aggravated at major referral centers like the Cleveland Clinic, in which many patients are sick enough to require urgent surgery. We therefore restricted our analysis to patients having strictly elective procedures, a design feature that probably accounts for our observed comparable mortality at various times of the day.

There was no statistically significant or clinically important association between day of the workweek and mortality. As with cardiac surgery,¹⁴ there thus seems to be no advantage in operating on any particular workday. Importantly though, neither study evaluated weekend surgery, which may be associated with more complications related to vascular surgery and obstetrical care (although perhaps fewer anesthetic complications).¹⁸ Mortality after intensive

care unit admission, adjusted for case mix, was similar on weekdays and weekend, and during the day and night, in one study.¹⁹ In another study, patients having nonemergency surgery, who were admitted to a regular nursing floor, had a higher 30-day mortality rate when surgery was performed on Friday than when surgery was performed on a Monday, Tuesday, or Wednesday.²⁰ However, the authors did not observe an increase in 30-day mortality in either outpatients or patients who were admitted to the intensive care unit, suggesting that the result may not be robust. Overall, available results suggest that day of the week is probably not a critical determinant of outcome for either elective surgical patients or critical care admissions.

There was also no association between month of the year and surgical mortality. Although an increase in adverse events in initial months of each training year, independent of level of training, has been reported,¹¹ our result is consistent with a previous study showing no increase in patient mortality in July in teaching hospitals.²¹ If there is a "July effect" for general surgery, it is probably small and of marginal clinical importance. Safety was maintained in July and August, despite incremental responsibility among trainees, presumably because of adequate (increased) vigilance on the part of more experienced physicians.

The relationship between moon phase and medical events and outcomes is a well-established "urban legend" that is even supported by the occasional study.^{22,23} As might be expected, we did not find that the phase of the moon had a significant association with 30-day mortality. This result is consistent with our previous work¹⁴ and most other studies.^{24–27}

Our study extends previous work by evaluating major complications, as defined by the AHRQ, along with mortality. Complications, which are common, are presumably more sensitive measures of outcome than death. Nonetheless, there were no statistically significant or clinically important time-dependent effects on major morbidity. This result strongly supports our main conclusion that time, day, and month of surgery have no substantial effect on the safety of elective surgery.

Our study was observational because practical considerations prevent randomizing enough patients to operation time, day, month, and moon phase to provide sufficient power to analyze events such as 30-day mortality, which have an incidence <1%. An unavoidable limitation of observational trials is poor protection against selection bias, measurement bias, and confounding. To minimize error, our results were risk-adjusted for demographic characteristics and various disease diagnoses; we also used a novel and powerful RSI¹² based on ICD-9 codes that are objectively defined and were carefully recorded during hospitalization (data acquisition for the registry is prospective, although our analysis was retrospective). Nonetheless, there are surely important prognostic factors that were unavailable to us. For example, sicker patients may have been systematically scheduled later in the day, although there is no evidence or even clinical impression to support this theory.

Emergency operations are generally riskier than their elective counterparts and are often performed on patients

in worse underlying medical condition. It is thus unsurprising that adverse outcomes, including mortality, are worse after emergency surgery. That emergency surgery is often required at night or on weekends compounds the problem, because the most skilled teams are often unavailable; this may explain why outcomes from cardiac arrest are worse at night and on weekends.²⁸ We thus considered only elective surgery. A consequence is that our study tested the effects of timing only during the routine workday and workweek. Even elective surgery performed late at night or on weekends may have considerably worse outcomes than those observed in our patients.¹⁶

Provider performance is not the only time-dependent factor potentially influencing outcome. Patients themselves demonstrate a circadian variation in numerous functions,^{29–31} although the extent to which circadian variation influences major outcomes remains unclear.³² However, the observed overall lack of time-dependence suggests that neither provider fatigue nor patient circadian rhythms is an important clinical determinant of major patient outcomes.

Mortality is the most important perioperative outcome but is rare (0.4% of our 32,001 patients) and thus less sensitive than other potential outcomes. It thus remains possible that there were important time-dependent outcomes that were not evaluated in this analysis.

The extent to which our results can be generalized to other teaching hospitals is an important question. The Cleveland Clinic is a tertiary referral center that performs high-risk surgery in a high-risk population. Mortality will presumably be less at institutions with a more typical (less acute) surgical population. However, there is no reason to assume that baseline or procedural risk per se influences the time-dependence of adverse outcomes. Because our results are risk-adjusted, they cannot be explained by local scheduling, such as performing the most complicated cases in the morning when personnel are fresh.

Resident work hours at the Clinic are limited, but no more so than required by the Residency Review Committee. Attending anesthesiologists mostly start at approximately 07:00, being relieved if necessary at some point in the late afternoon. But a few clinicians start late and stay late, and others are on night call. Surgeons usually complete their scheduled lists. There is also a limited nursing shift change at 15:00, but this sort of staffing pattern is typical in most teaching hospitals.

The results of this study are nonetheless a function of many attributes of the Cleveland Clinic adult general specialty operating rooms, including its case mix, staffing, and other factors. Other hospitals will need to evaluate the similarity of their case mix, staffing, and other attributes before drawing conclusions about the possible applicability of our study's findings to their hospital. Importantly, this study explicitly excluded all urgent and emergent cases, which might be especially affected by performance-shaping factors, including fatigue and circadian effects.

In summary, there were also no significant time-dependent differences in mortality or composite complications. Elective surgery thus appears to be comparably safe at any time of the workday, any day of the workweek, and in any month of the year in our teaching hospital. ■■

DISCLOSURES

Name: Daniel I. Sessler, MD.

Contribution: Daniel I. Sessler helped write the manuscript.

Attestation: Daniel I. Sessler has seen the original study data, reviewed the analysis of the data, approved the final manuscript, and is the author responsible for archiving the study files.

Name: Andrea Kurz, MD.

Contribution: Andrea Kurz helped design the study, conduct the study, and write the manuscript.

Attestation: Andrea Kurz has seen the original study data, reviewed the analysis of the data, and approved the final manuscript.

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Contribution: Leif Saager helped design the study, conduct the study, and write the manuscript.

Attestation: Leif Saager has seen the original study data, reviewed the analysis of the data and approved the final manuscript.

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Attestation: Jarrod E. Dalton has seen the original study data, reviewed the analysis of the data, and approved the final manuscript.

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REFERENCES

- Gaba DM, Howard SK. Patient safety: fatigue among clinicians and the safety of patients. *N Engl J Med* 2002;347:1249–55
- Mitler MM, Carskadon MA, Czeisler CA, Dement WC, Dinges DF, Graeber RC. Catastrophes, sleep, and public policy: consensus report. *Sleep* 1988;11:100–9
- Fisman DN, Harris AD, Rubin M, Sorock GS, Mittleman MA. Fatigue increases the risk of injury from sharp devices in medical trainees: results from a case-crossover study. *Infect control Hosp Epidemiol* 2007;28:10–7
- Nasca TJ, Day SH, Amis ES Jr. The new recommendations on duty hours from the ACGME Task Force. *N Engl J Med* 2010;363:e3
- Kuhn G. Circadian rhythm, shift work, and emergency medicine. *Ann Emerg Med* 2001;37:88–98
- Smith-Coggins R, Rosekind MR, Hurd S, Buccino KR. Relationship of day versus night sleep to physician performance and mood. *Ann Emerg Med* 1994;24:928–34
- Weinger MB, Englund CE. Ergonomic and human factors affecting anesthetic vigilance and monitoring performance in the operating room environment. *Anesthesiology* 1990;73:995–1021
- Owens JA. Sleep loss and fatigue in healthcare professionals. *J Perinat Neonatal Nurs* 2007;21:92–100
- Ascher-Walsh CJ, Capes T. An evaluation of the resident learning curve in performing laparoscopic supracervical hysterectomies as compared with patient outcome: five-year experience. *J Minim Invasive Gynecol* 2007;14:719–23
- Siqueira TM Jr, Gardner TA, Kuo RL, Paterson RF, Stevens LH, Lingeman JE, Shalhav AL. One versus two proficient laparoscopic surgeons for laparoscopic live donor nephrectomy. *Urology* 2002;60:406–9
- Haller GSA, Myles PS, Taffé T, Perneger TV, Wu CC. Increase in the rate of undesirable events at the beginning of the academic year (abstract). *Anesthesiology* 2009;111:A23
- Sessler DI, Sigl JC, Manberg PJ, Kelley SD, Schubert A, Chamoun N. A broadly applicable risk stratification system for predicting duration of hospitalization and mortality. *Anesthesiology* 2010;113:1026–37
- Harrell FE. Regression modeling strategies: with applications to linear models, logistic regression, and survival analysis. New York: Springer, 2001
- Tan PJ, Xu M, Sessler DI, Bashour CA. Operation timing does not affect outcome after coronary artery bypass graft surgery. *Anesthesiology* 2009;111:785–9
- George TJ, Arnaoutakis GJ, Merlo CA, Kemp CD, Baumgartner WA, Conte JV, Shah AS. Association of operative time of day with outcomes after thoracic organ transplant. *JAMA* 2011;305:2193–9
- Kelz R. Time of day is associated with postoperative morbidity: an analysis of the National Surgical Quality Improvement Program data. *Ann Surg* 2008;247:544–52
- Wright MC, Phillips-Bute B, Mark JB, Stafford-Smith M, Grichnik KP, Andregg BC, Taekman JM. Time of day effects on the incidence of anesthetic adverse events. *Qual Saf Health Care* 2006;15:258–63
- Bendavid E, Kaganova Y, Needleman J, Gruenberg L, Weissman JS. Complication rates on weekends and weekdays in US hospitals. *Am J Med* 2007;120:422–8
- Wunsch H, Mapstone J, Brady T, Hanks R, Rowan K. Hospital mortality associated with day and time of admission to intensive care units. *Intensive Care Med* 2004;30:895–901
- Zare MM, Itani KM, Schiffner TL, Henderson WG, Khuri SF. Mortality after nonemergent major surgery performed on Friday versus Monday through Wednesday. *Ann Surg* 2007;246:866–74
- Englesbe MJ, Fan Z, Baser O, Birkmeyer JD. Mortality in medicare patients undergoing surgery in July in teaching hospitals. *Ann Surg* 2009;249:871–6
- Wells RJ, Gionfriddo JR, Hackett TB, Radecki SV. Canine and feline emergency room visits and the lunar cycle: 11,940 cases (1992–2002). *J Am Vet Med Assoc* 2007;231:251–3
- Ali Y, Rahme R, Matar N, Ibrahim I, Menassa-Moussa L, Maarrawi J, Rizk T, Nohra G, Okais N, Samaha E, Moussa R. Impact of the lunar cycle on the incidence of intracranial aneurysm rupture: myth or reality? *Clin Neurol Neurosurg* 2008;110:462–5
- Das S. Do lunar phases affect conception rates in assisted reproduction? *J Assist Reprod Genet* 2005;22:15–8
- Biermann T, Estel D, Sperling W, Bleich S, Kornhuber J, Reulbach U. Influence of lunar phases on suicide: the end of a myth? A population-based study. *Chronobiol Int* 2005;22:1137–43
- Kredel M, Goepfert C, Bassi D, Roewer N, Apfel CC. The influence of the weather and the phase of the moon on post-operative nausea and vomiting. *Acta Anaesthesiol Scand* 2006;50:488–94
- Holzheimer RG, Nitz C, Gresser U. Lunar phase does not influence surgical quality. *Eur J Med Res* 2003;8:414–8
- Peberdy MA, Ornato JP, Larkin GL, Braithwaite RS, Kashner TM, Carey SM, Meaney PA, Cen L, Nadkarni VM, Praestgaard AH, Berg RA. Survival from in-hospital cardiac arrest during nights and weekends. *JAMA* 2008;299:785–92
- Wright KP Jr, Hull JT, Czeisler CA. Relationship between alertness, performance, and body temperature in humans. *Am J Physiol Regul Integr Comp Physiol* 2002;283:R1370–7
- Tayefeh F, Plattner O, Sessler DI, Ikeda T, Marder D. Circadian changes in the sweating-to-vasoconstriction interthreshold range. *Pflügers Arch* 1998;435:402–6
- Sessler DI, Lee KA, McGuire J. Isoflurane anesthesia and circadian temperature cycles. *Anesthesiology* 1991;75:985–9
- Klerman EB. Clinical aspects of human circadian rhythms. *J Biol Rhythms* 2005;20:375–86